# SOC Lab Setup Report

**Name:** Thato
**Programme:**Cybersecurity internship
**Task:** Day 1 – Build a SOC Lab
**Date:** 21/09/2026

## 1. Objective

The goal of this task was to build a small, isolated Security Operations Centre (SOC) lab. The lab pairs an attacker/analyst machine (Linux) with a monitored endpoint (Windows) on a private network. It will serve as the base for log collection, SIEM work and incident investigations later in the programme. An isolated lab lets me generate and study suspicious activity safely, without risking my host machine or any real network.

## 2. Lab Environment

| Component | Details |
|---|---|
| Host machine | [OS, CPU, RAM, free disk] |
| Hypervisor | VirtualBox [version 13] |
| Linux VM | [Kali Linux ], [2] vCPU , [4] GB RAM, [64] GB disk |
| Windows VM | [Windows 10 version], [4] vCPU, [8] GB RAM, [256] GB disk |
| Lab network | VirtualBox Internal Network named `[soclab]`, subnet [192.168.56.0/24] |
| Linux IP | [192.168.56.10] |
| Windows IP | [192.168.56.20] |

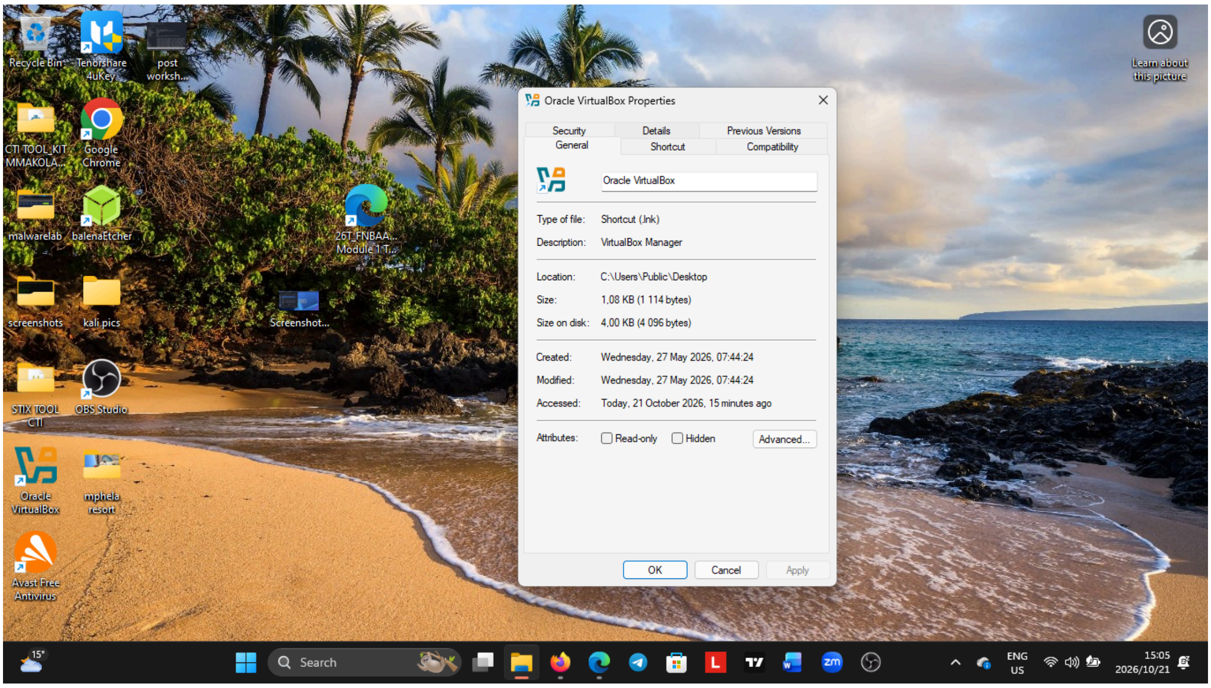
| | |
|---|---|
| Internet access | [NAT on Adapter 2 for updates / none] |

# 3. Setup Steps

## 3.1 Installing VirtualBox

I went to to duckduckgo and downloaded a virtualbox from [oracle.org]() and installed it by following prompts

*Screenshot 1:*



Oracle VirtualBox Properties

Oracle VirtualBox

Type of file: Shortcut (.lnk)
Description: VirtualBox Manager
Location: C:\Users\Public\Desktop
Size: 1,08 KB (1 114 bytes)
Size on disk: 4,00 KB (4 096 bytes)
Created: Wednesday, 27 May 2026, 07:44:24
Modified: Wednesday, 27 May 2026, 07:44:24
Accessed: Today, 21 October 2026, 15 minutes ago
Attributes: ☐ Read-only ☐ Hidden

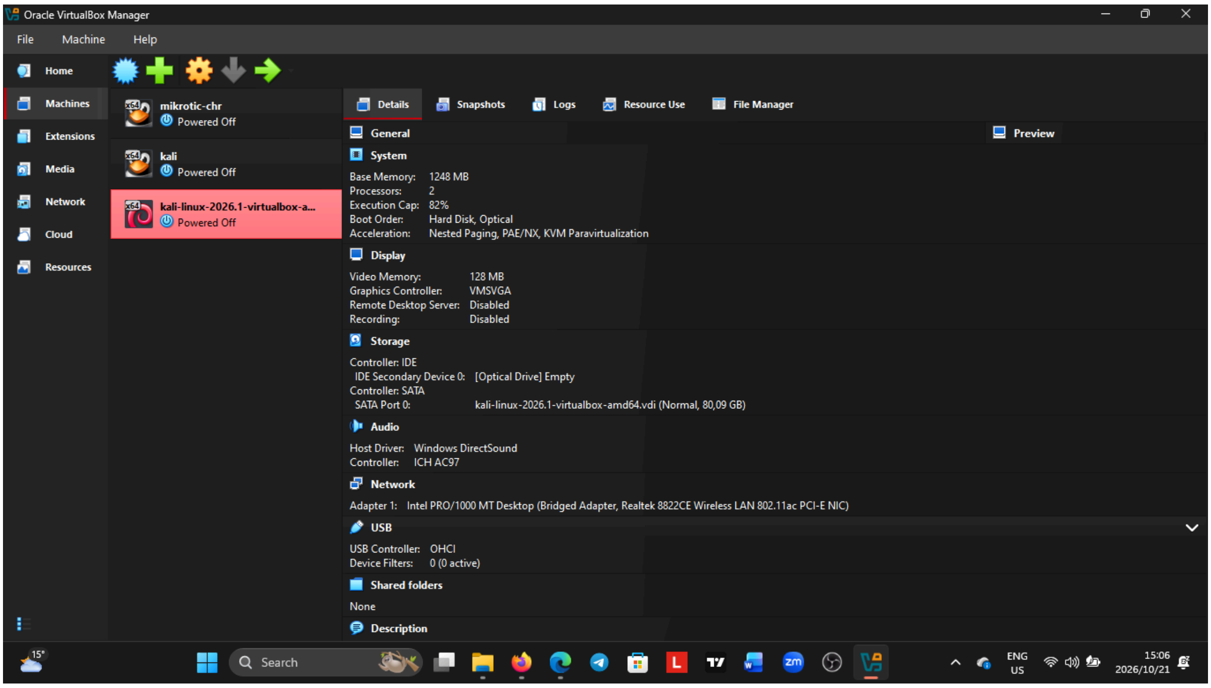
Security
Details
Previous Versions
General
Shortcut
Compatibility
Advanced...
OK
Cancel
Apply
Recycle Bin
Tenorshare 4uKey
post worksh...
CTI TOOL KIT MMAKOLA...
Google Chrome
malwarelab
balenaEtcher
26T FNBAA... Module 1 T...
screenshots
kali pics
Screenshot...
STIX TOOL CTI
OBS Studio
Oracle VirtualBox
mphela resort
Avast Free Antivirus
Learn about this picture
15°
Search
ENG US
15:05 2026/10/21

*VirtualBox Manager home screen / version.*

## 3.2 Linux VM

After i installed virtualbox from oracle, i went on and downloaded kali iso image from kali, and unzipped the file with 7z(zip) then added my iso in tto my vm and added required cpu, ram , and network settings then i set my username and password then tested , and good thing it works just fine

*Screenshot 2:*


Oracle VirtualBox Manager
File
Machine
Help
Home
Machines
Extensions
Media
Network
Cloud
Resources
mikrotic-chr
Powered Off
kali
Powered Off
kali-linux-2026.1-virtualbox-a...
Powered Off
Details
Snapshots
Logs
Resource Use
File Manager
Preview
Search
ENG
US
15:06
2026/10/21

**General**

**System**

Base Memory: 1248 MB
Processors: 2
Execution Cap: 82%
Boot Order: Hard Disk, Optical
Acceleration: Nested Paging, PAE/NX, KVM Paravirtualization

**Display**

Video Memory: 128 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled

**Storage**

Controller: IDE
IDE Secondary Device 0: [Optical Drive] Empty
Controller: SATA
SATA Port 0: kali-linux-2026.1-virtualbox-amd64.vdi (Normal, 80,09 GB)

**Audio**

Host Driver: Windows DirectSound
Controller: ICH AC97

**Network**

Adapter 1: Intel PRO/1000 MT Desktop (Bridged Adapter, Realtek 8822CE Wireless LAN 802.11ac PCI-E NIC)

**USB**

USB Controller: OHCI
Device Filters: 0 (0 active)

**Shared folders**

None

**Description**

*Linux VM running (desktop or terminal).*

## 3.3 Windows VM

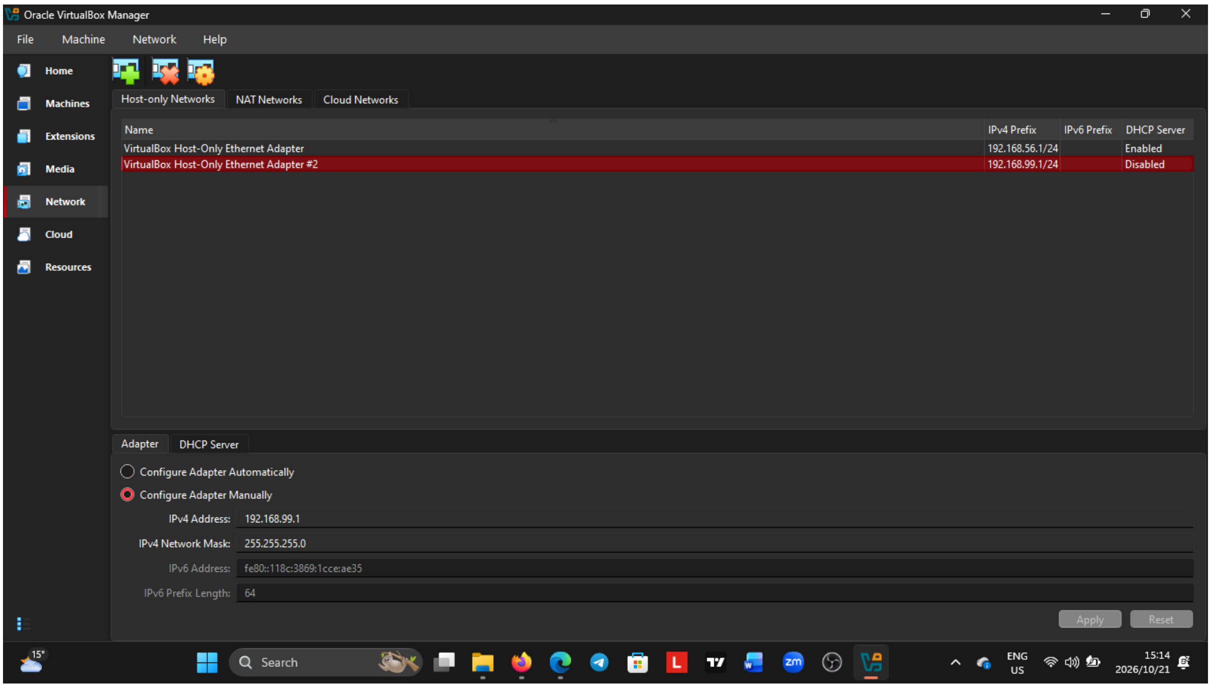
Here i went to microsoft and downloaded windows 10 iso and added it to my vm machine by following prompts

*Screenshot 3: will be aaded at the end .*

## 3.4 Network configuration

both VMs were set to Internal Network `soclab` under Settings → Network, and i assigned static IPs on each machine.]

*Screenshot 4:*

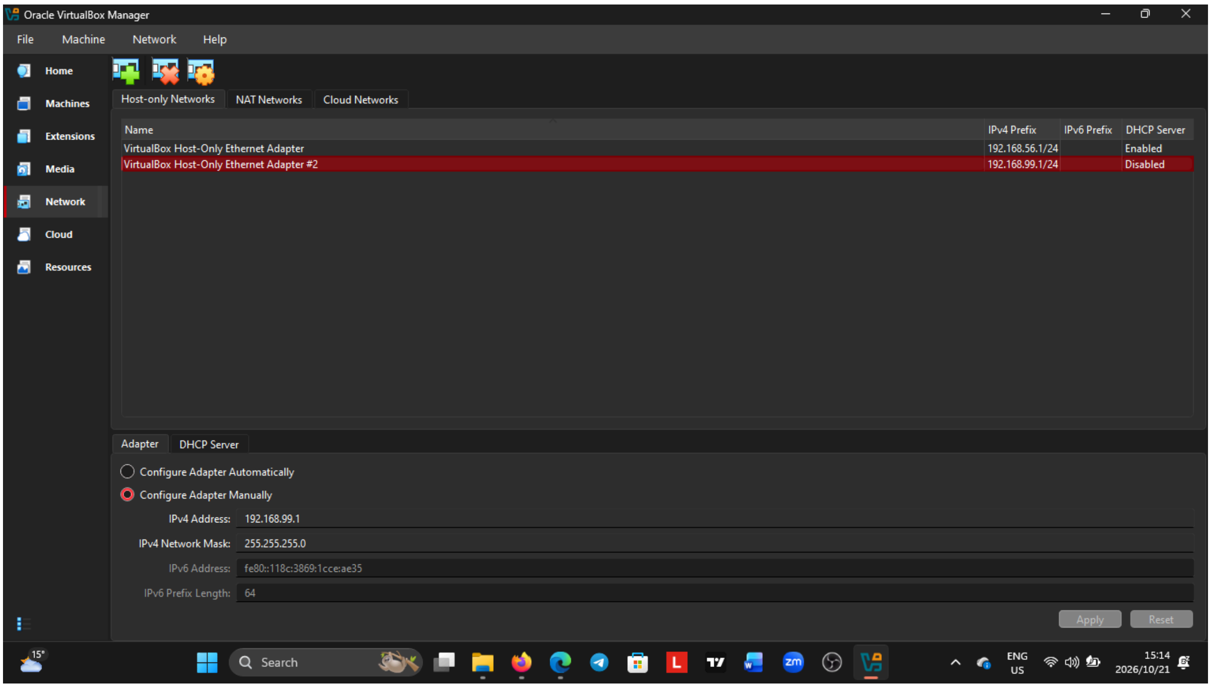

Oracle VirtualBox Manager
File
Machine
Network
Help
Home
Machines
Extensions
Media
Network
Cloud
Resources
Host-only Networks
NAT Networks
Cloud Networks
Adapter
DHCP Server
Apply
Reset
Search
15°
ENG
US
15:14
2026/10/21

| Name | IPv4 Prefix | IPv6 Prefix | DHCP Server |
|---|---|---|---|
| VirtualBox Host-Only Ethernet Adapter | 192.168.56.1/24 | | Enabled |
| VirtualBox Host-Only Ethernet Adapter #2 | 192.168.99.1/24 | | Disabled |

○ Configure Adapter Automatically
● Configure Adapter Manually
IPv4 Address: 192.168.99.1
IPv4 Network Mask: 255.255.255.0
IPv6 Address: fe80::118c:3869:1cce:ae35
IPv6 Prefix Length: 64

*Network settings of the Linux VM.*

*Screenshot 5:*


Oracle VirtualBox Manager
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Search
15°
ENG
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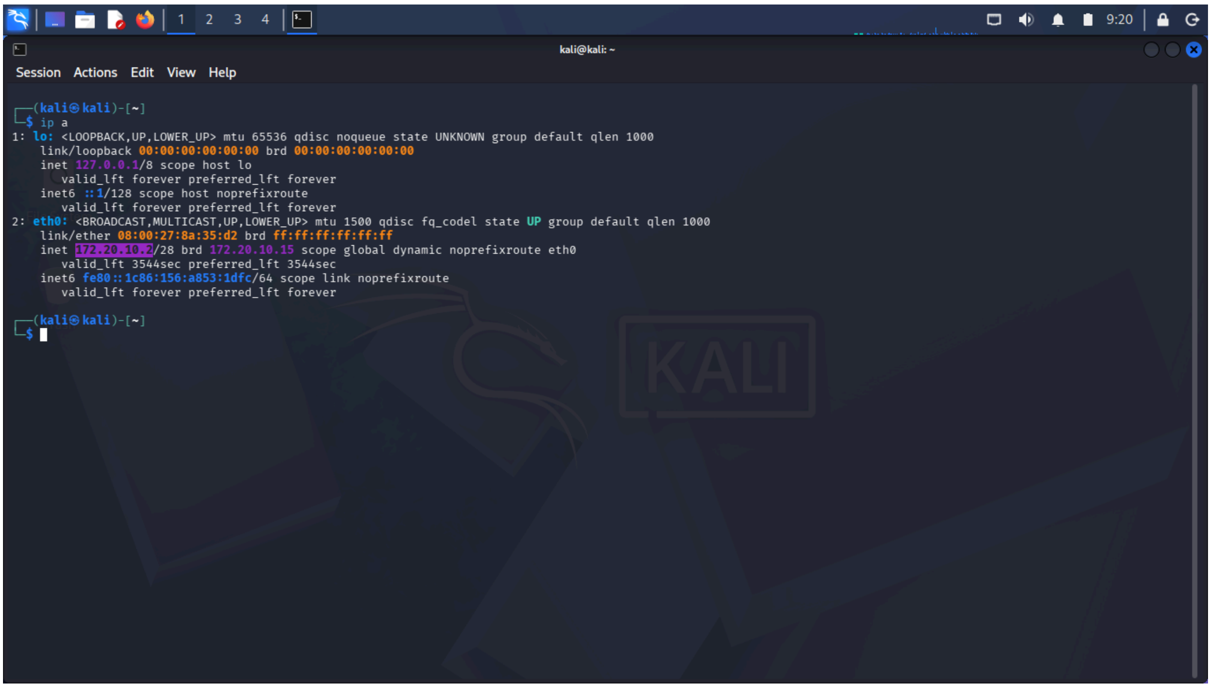
*Network settings of the Windows VM.*

# 4. Connectivity Testing

| **Test** | **Command** | **Result** |
|---|---|---|

| | | |
|---|---|---|
| Linux IP check | `ip a` | [IP shown 192.168.99.1 ] |
| Windows IP check | `ipconfig` | [IP shown 192.168.99.1] |
| Linux → Windows | `ping [Windows IP]` | [Success] |
| Windows → Linux | `ping [Linux IP]` | [Success] |
| Host discovery | `nmap -sn [subnet]` | [Both hosts discovered] |

*Screenshot 6:*



```
┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 172.20.10.2/28 brd 172.20.10.15 scope global dynamic noprefixroute eth0
       valid_lft 3544sec preferred_lft 3544sec
    inet6 fe80::1c86:156:a853:1dfc/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(kali㉿kali)-[~]
└─$ 
```

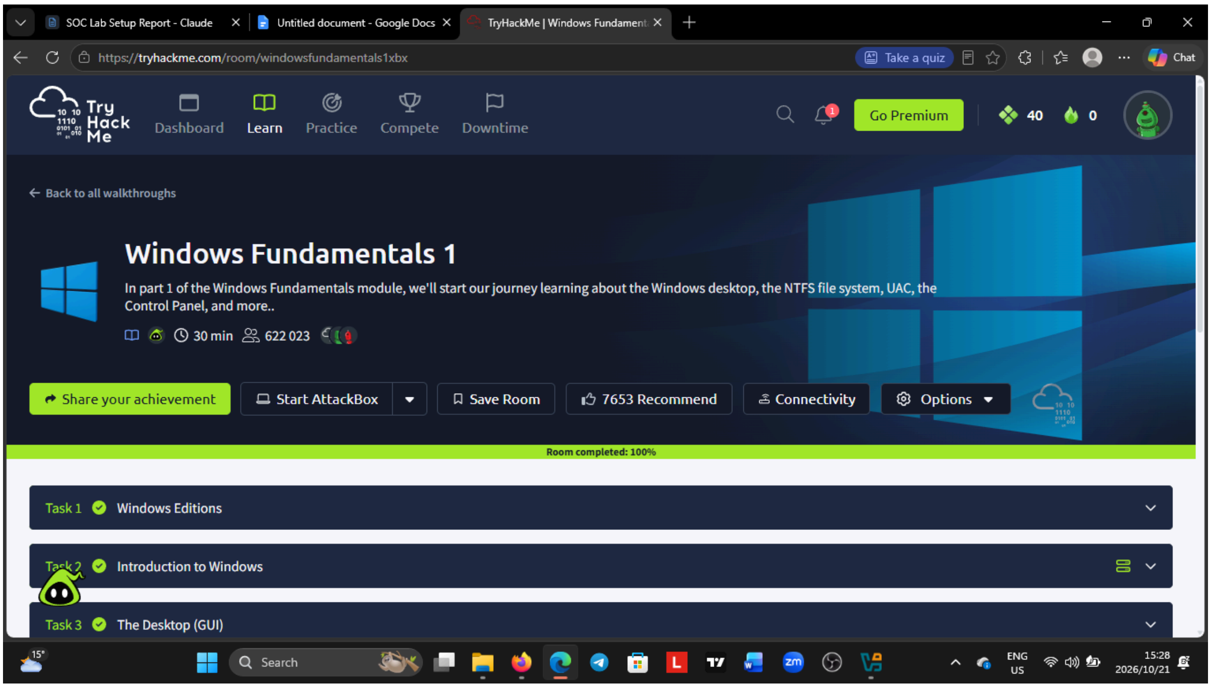
*`ip a` on Linux and `ipconfig` on Windows.*

## 5. Problems Encountered and Fixes

At first i didnt know difference between NAT and bridged only so it gave me problems because network connection was falling i had to learn about them over and over again with the help of google

## 6. TryHackMe: Windows Fundamentals 1

I completed the free room *Windows Fundamentals 1*.



Windows Fundamentals 1

In part 1 of the Windows Fundamentals module, we'll start our journey learning about the Windows desktop, the NTFS file system, UAC, the Control Panel, and more..

Room completed: 100%

Task 1 Windows Editions

Task 2 Introduction to Windows

Task 3 The Desktop (GUI)

SOC Lab Setup Report - Claude
Untitled document - Google Docs
TryHackMe | Windows Fundament
https://tryhackme.com/room/windowsfundamentals1xbx
Take a quiz
Chat
Try Hack Me
Dashboard
Learn
Practice
Compete
Downtime
Go Premium
40
0
Back to all walkthroughs
30 min
622 023
Share your achievement
Start AttackBox
Save Room
7653 Recommend
Connectivity
Options
Search
ENG
US
15:28
2026/10/21

Key things I learned user account control and task manager and system settings and these matter for a SOC analyst because majority of corporate networks run on windows and to catch bad guys you must know how system is supposed to look normaly

## 8. Conclusion

Now i can deploy , managev , test vm within sandboxed environment through this process and i alsp learned that using network isolation keeps testing safe from external threats